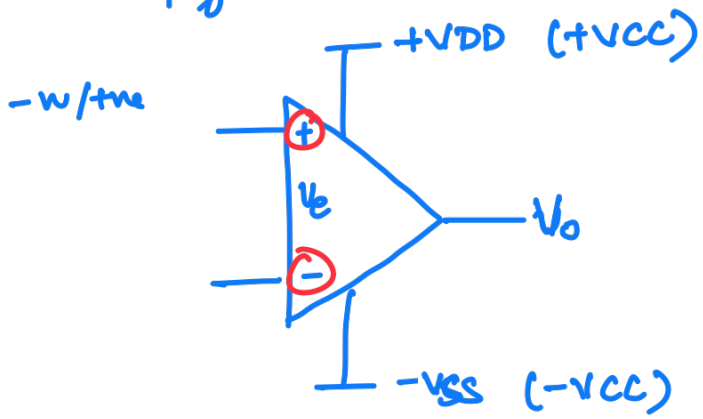
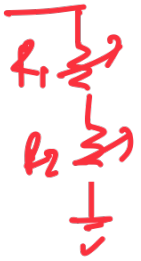
# Lecture #03

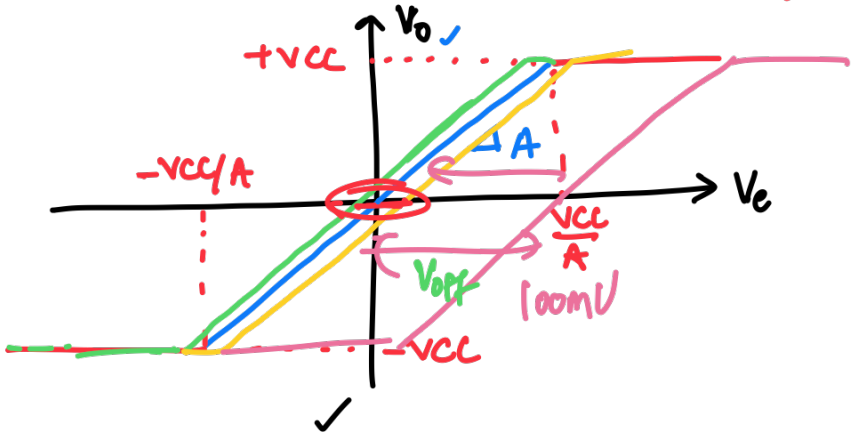
## Opamp (Operational Amplifier)



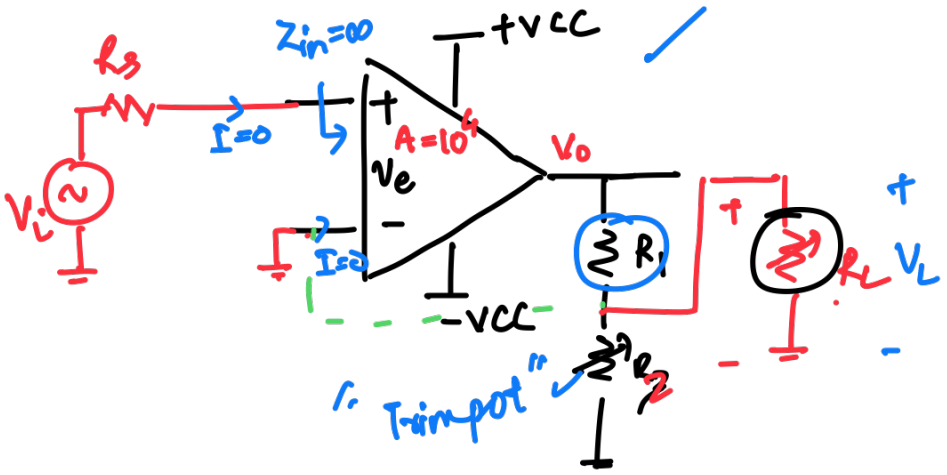




$V_e = V_+ - V_-$

$V_o = A V_e$




- Gain A : 10 - $10^6$







A = k = 75

k = 101 = A

k = 220

200 - 250

10000

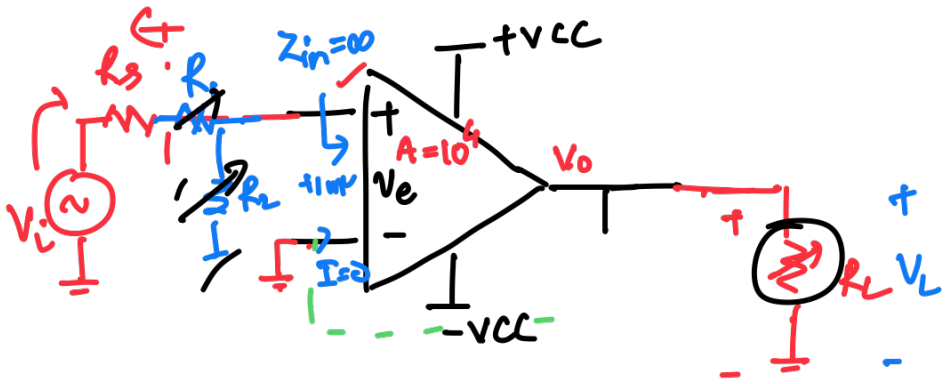
$V_o = A V_e$

$V_e = V_+ - V_- = V_i - 0 = V_i$

$V_o = A V_e = A V_i$

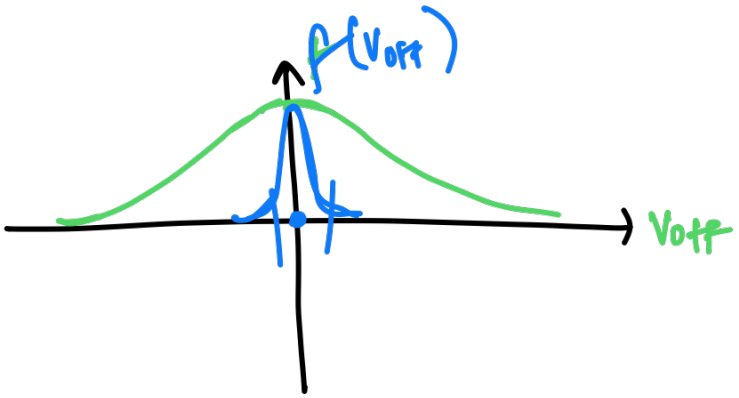
$V_i = \frac{(R_2 \| R_L)}{(R_1 \| R_2) + R_1} V_o = \frac{(R_2 \| R_L)}{(R_1 \| R_2) + R_1} A \cdot V_i$

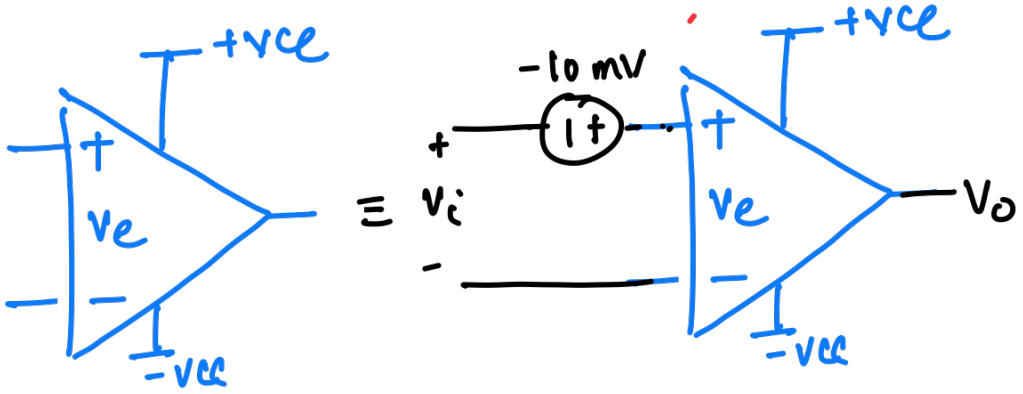
$k = 220$




$$V_e = \frac{R_2}{R_2 + R_1 + R_3} \cdot V_i$$

$V_o = A V_e$



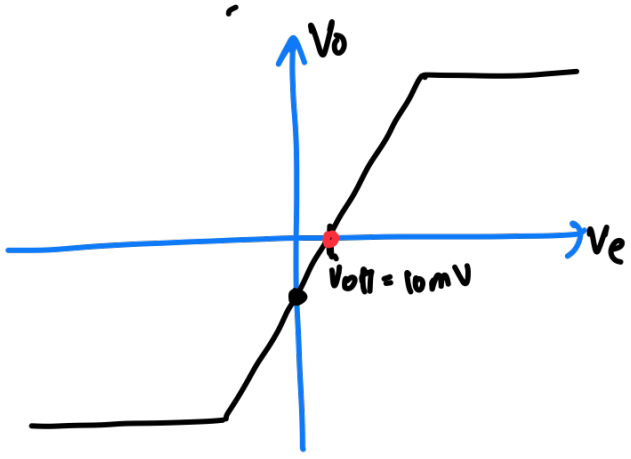




$A = 10000$

$V_{off} = 10mV$

$A = 10000$

$V_{off} = 0mV$




$V_i = 0$, $V_e = -10mV$, $V_o = A \times -10mV$

$V_i = +10mV$, $V_e = 0$, $V_o = A \times 0 = 0$

- $V_o = AV_e$

$= A_1 V_e + A_2 V_e^2 + A_3 V_e^3 + \ldots$

$A_1$ = 1000

$\sin(3\omega_i t) = 3\sin(\omega_i t) - 4\sin^3(\omega_i t)$

$\sin^3(\omega_i t) = \frac{3\sin(\omega_i t)}{4} - \frac{\sin(3\omega_i t)}{4}$

$V_e = V_i$

$V_i = a\sin(\omega_i t) = V_e$

$V_e^2 = a^2\sin^2(\omega_i t) = \frac{a^2}{2}(1 - \cos(2\omega_i t))$

$V_e^3 = a^3\sin^3(\omega_i t) = a^3\left(\frac{3\sin(\omega_i t)}{4} - \frac{\sin(3\omega_i t)}{4}\right)$

